00-D-1718
May 9, 1972

FEDERAL SPECIFICATION

DISPENSER, DETERGENT AND RINSE ADDITIVE, CENTRAL SYSTEM LIQUID TYPE FOR COMMERCIAL DISHWASHING MACHINES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers remote units for automatically injecting a liquid detergent into the wash water and a liquid drying agent into the final rinse water of commercial dishwashing machines.

1.2 Classification. The dispensers shall be of the following types as specified (see 6.2).

Type I - For use with multiple product detergents.
Type II - For use with single product detergent.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

P-R-1272 - Rinse Additive, Dishwashing.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

National Sanitation Foundation (NSF):

Listing of Food Service Equipment.

Standard No. 29 - Detergent and Chemical Feeders for Commercial Spray Type Dishwashing Machines.

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Underwriters' Laboratories, Inc. (UL) Standard:

Standard No. 921 - Commercial Electric Dishwashers.

(Application for copies should be addressed to the Underwriters' Laboratories Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; or 1655 Scott Boulevard, Santa Clara, California 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2 and 6.3).

3.2 Standard product. The dispenser delivered under this specification shall be the manufacturer's current standard commercial product, except for any changes necessary to comply with specification requirements.

3.3 Standards.

3.3.1 UL standards. The dispenser shall conform to the applicable requirements of UL Standard 921.

3.3.1.1 UL certification. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dispensers he proposes to furnish under this specification meet the applicable requirements of UL Standard 921. Acceptable evidence of meeting the applicable requirements of UL standards shall be the UL label, or listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the dispenser offered has been tested and conform to the requirements of UL standards as applicable.

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3.3.2 NSF standard. Dispensers shall conform to the applicable requirements of NSF Standard No. 29.

3.3.2.1 NSF certification. The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the dispensers he proposes to furnish under this specification meets the applicable requirements of NSF Standard No. 29 as follows:

(1) Listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished dispenser shall be submitted prior to approval of the first article, if one is submitted, or prior to approval of the first shipment, or

(2) A certification for the dispenser issued by NSF under their special one-time evaluation/certification service shall be submitted prior to approval of the first article, if one is submitted, or prior to approval of the first shipment, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the dispensers are being procured, indicating that the dispenser has been tested and conforms to NSF Standard No. 29 shall be submitted prior to the approval of the first article, if one is submitted, or prior to commencing production.

3.4 Design. The dispenser shall be an electro-mechanical system capable of dispensing and maintaining from a remote location (see 6.2) a concentration of liquid detergent within a range of **0.10 to 0.40 percent in the wash** water; and a rinse additive in the final rinse water for commercial dishwashing machines. The system shall operate satisfactorily when tested in accordance with 4.4.1 and 4.4.2. An electronic sensing device shall be provided for mounting on the dishwasher for monitoring the dispensing of the detergent. Detergent and rinse additive bulk containers shall be a mimimum 14 gallons each. The system shall contain a failure alarm system (light and audible signals) to indicate equipment malfunction. The system shall also contain a seperate alarm to indicate when the detergent and rinse additive bulk containers need replenishing.

3.4.1 Master panel. The master panel shall be remote from the dishwasher machine as specified in 6.2 and shall contain controls, and components necessary for adjusting and maintaining concentration levels of detergent, and rinse additive specified. The panel shall contain the failure light specified in 3.4.

3.4.2 Sub-panel 1. A sub-panel shall be located on the dishwashing machine and contain failure alarm indicated in 3.4.

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3.4.3 Sub-panel 2. When specified (see 6.2), a sub-panel shall be provided with failure alarm specified in 3.4 where supervisory personnel can observe when detergent and rinse additive bulk containers need servicing or when a malfunction occurs.

3.4.4 Test kit. A test kit shall be provided containing 4 ozs. of methylorange indicator solution, 4 ozs. of 0.5N sulfuric acid and one standard medicine dropper.

3.4.5 Electrical characteristics. Unless otherwise specified (see 6.2), the dispenser system shall be designed for operation on a nominal 120 volt, 60 hertz, 1-phase electrical system.

3.5 Marking for identification. The manufacturer's name, contract number, model number, and serial number shall be permanently marked on a corrosion **resisting** plate located on or in the master panel.

3.6 Operating instructions. Instructions indicating the proper operating procedures for using the system shall be etched, stamped, or embossed on corrosion **resisting** metal or plastic plate and permanently attached in a visable position for the operator.

3.7 Workmanship. Components shall be finished with all burrs and rough edges removed. Welds shall be sound, smooth, and free from pits, fractures, burn-through, or fissures. Spot welds shall be free from pits and flashes. Bolt holes shall be accurately punched or drilled and shall have burrs removed. Washers and lockwashers shall be provided in accordance with standard commercial practice, and all bolts, nuts, and screws shall be drawn tight. Rivets shall be tightly driven, fully peened, and shall completely fill the holes. Electrical wiring shall be provided with adequate slack to relieve strain and the wiring shall not be abraded, poorly joined or loose at terminals. Plumbing shall be installed in accordance with standard plumbing code and practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article is required, it shall be examined for defects listed in table I and tested as specified in 4.4.1 and 4.4.2. Failure to pass any tests shall be cause for rejection of

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the first article. The presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the first article.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 End item inspection. The inspection lot shall consist of all dispensers offered for inspection at one time. The sample unit shall be one completely fabricated dispenser.

4.3.2.1 Visual examination. The dispenser shall be examined for defects in table I. The inspection shall be level II with and acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examination	Defect	Classification Major	Minor
Consturction and workmanship	Any component missing or not as specified	x	
	Any component fractured or malformed	x	
	Any burr or rough edge	x	
	Functioning component that requires abnormal force to operate	x	
	Any adjustable assembly not adjusted to perform the function intended		x
	Any component not accessible for replacement or repair	x	
Welding	Any weld fractured, not sound, or contains fissures or burn-through	x	

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TABLE I. Classification of defects (cont'd)

Examination	Defect	Classification Major	Minor
Welding (cont'd)	Any weld not smooth or contains pits		x
	Spot weld contains flashes		x
Bolts, nuts, screws	Stripped or fractured	x	
	Loose or not provided with lockwashers		x
Rivets	Loose, not fully peened, or fractured	x	
Electrical assemblies (when applicable)	Wiring cut, abraded, or loosely joined at terminals	x	
	Adequate slack not provided for wiring to relieve strain	x	
	Insulation stripped from wire	x	
Identification marking	Missing, incomplete, not legible, not located as specified		x

4.3.2.2 Capacity examination. Capacity examination shall be made of the bulk containers to determine compliance with 3.4. Any reservoir capacity that is not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.3 Testing of the end item. When a first article is not required, the initial production unit shall be tested as specified in 4.4.1 and 4.4.2. In addition, one unit from each production lot shall be tested as specified in 4.4.1 and 4.4.2. Failure to pass any test shall be cause for rejection of the lot.

4.3.4 Standards compliance. Proof of compliance with the applicable requirements of 3.3.1 and 3.3.2 shall be submitted to the Government representative.

4.3.5 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The sample unit shall be one shipping container

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fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-4 with an AQL of 6.5 defects expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examination	Defect
Markings	Omitted; incorrect; illegible, or improper size; location, sequence, or method of application.
Materials	Components missing, damaged, or otherwise defective.

4.3.5.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.4 Tests.

4.4.1 Detergent dispenser operational test. A two or three tank commercial dishwasher and a colorless detergent shall be used for the test. Fill the wash tank with clean **150° to 160°F water to the overflow level.** Turn off the water inlet valves. Adjust the detergent dispenser to provide a 0.10 percent wash tank solution. Turn on the dishwashing machine to circulate the water. In accordance with operating instructions, start the dispenser system in operation and allow it to feed until it ceases dispensing, showing initial feeding has been completed in accordance with the preset detergent concentration. At this point, remove a water sample and test for concentration using the test kit (see 4.4.1.1). Record all readings taken during this test. All samples of wash solution shall be drawn at the moment the dispenser begins to feed after a period of holding at OK. In titrating the wash solution, the titrating solution must be added with the dropper held vertically in order to insure drops of uniform size. It is essential that any sample drawn for titration be identical in concentration to the solution surrounding the electrode. Be sure that samples are not taken during or immediately following a period when detergent is being discharged into the wash tank. Turn on the inlet valve and allow water to enter the wash tank at a rate of 2 quarts ± 1/2 pint per minute. Continue testing water samples as specified above for 1-hour during which water samples shall be tested for concentration every 10 minutes to determine if the dispenser is capable of maintaining the concentration specified above. Test alarm system with an overdose of detergent to determine conformance with 3.4. Prior to conclusion of the test, actuate the alarm system by simulating malfunction of the fill

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valve, drain valve, or rinse valve to determine conformance with 3.4. Inability of the dispensing unit to maintain the specified concentration above or the alarm system to activate for overdose or mechanical malfunction shall constitute failure of the test.

4.4.1.1 Test kit. For determining concentration, the following test apparatus and procedure shall be followed: Take a 10± 0.5 cc sample volume of wash water with detergent from the wash tank. Add two drops of methylorange indicator solution. Titrate 0.5 N sulfuric acid drops from a standard medicine dropper until a pink end-point, from orange to pink, is obtained. The following table may be used for determining concentrations.

Per cent Concentration 1/	Drops of 0.5 N Sulfuric Acid
0.10	4
.15	5
.20	7
.30	9
.40	12

1/ Interpolated readings may be made between these table readings.

4.4.2 Rinse additive dispenser operational test. A full load of flatware and tableware shall be subjected to the final rinse water containing rinse additive conforming to P-R-1272; and adjustment made for optimum sheeting. The inability of the rinse additive dispenser to dispense sufficient rinse additive needed for good sheeting shall constitute failure of the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The components of the dispenser unit shall be packaged in snug-fitting folding or set-up paperboard of fiberboard boxes conforming to variety 1 of PPP-B-566, variety 1 of PPP-B-676, or type CF, variety SW, or type SF, class domestic of PPP-B-636, respectively, except that the weight of contents of paperboard boxes shall not exceed 10 pounds. Components shall be cushioned as necessary to prevent movement or damage while in transit. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Level C. The dispenser units shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each complete dispenser unit, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with PPP-B-636.

5.2.2 Level B. Each complete dispenser unit, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fabricated and closed in accordance with PPP-B-636.

5.2.3 Level C. Dispenser units, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil Agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military Agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dispenser covered herein is intended for use with multiple tank high capacity commercial dishwashing machines.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of dispenser required (see 1.2).
(c) When a first article is required (see 3.1, 4.2, and 6.3).
(d) Location of master panel and bulk containers (see 3.4).

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(e) When sub-panel is required and distance from dispenser (see 3.4.3).
(f) Electrical characteristics if other than specified (see 3.4.5).
(g) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(h) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection, test and approval of the first article.

6.4 Liquid rinse additive for use with the rinse injector covered by this specification may be requistioned from GSA by referencing FSN 7930-685-4762 (1/2 gallon) or FSN 7930-619-9575 (1 gallon).

6.5 Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7320-0518

Review activity:

Army - MD

Civil Agency Coordinating Activity:

VA-DMS

User activity:

Army - CE

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION DISPENSER, DETERGENT AND RINSE ADDITIVE, CENTRAL SYSTEM LIQUID TYPE FOR COMMERCIAL DISHWASHING MACHINES 00-D-1718

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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